

2. The velocity- time graph of a particle moving in a straight line is shown in Fig. 2:

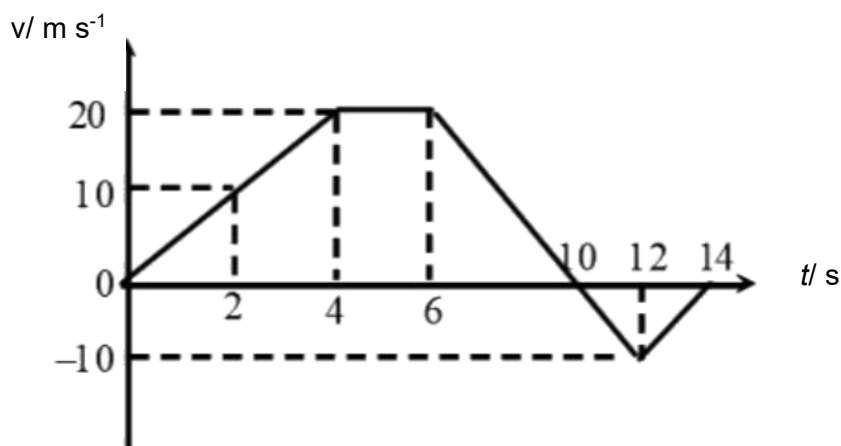


Fig. 2

What are the average speed and velocity of the particle between time $t = 0$ and $t = 14$ s?

	Average speed / m s^{-1}	Average velocity / m s^{-1}
A	7.1	7.1
B	7.1	10
C	10	7.1
D	10	10

Ans: C

Displacement = Area under velocity-time graph

$$\begin{aligned}
 &= \frac{1}{2}(2+10)(20) - \frac{1}{2}(10)(4) && \begin{array}{l} \text{Above time axis: positive displacement} \\ \text{Below time axis: negative displacement} \end{array} \\
 &= 120 - 20 \\
 &= 100 \text{ m}
 \end{aligned}$$

$$\begin{aligned}
 \text{Average velocity} &= \frac{\text{displacement}}{\text{time}} = \frac{100}{14} \\
 &= 7.1 \text{ m s}^{-1}
 \end{aligned}$$

Distance = Area under velocity-time graph

$$\begin{aligned}
 &= \frac{1}{2}(2+10)(20) + \frac{1}{2}(10)(4) \\
 &= 120 + 20 \\
 &= 140 \text{ m}
 \end{aligned}$$

Distance is a scalar and has no direction. So areas above and below time axis are considered as positive.

$$\begin{aligned}
 \text{Average velocity} &= \frac{\text{displacement}}{\text{time}} = \frac{140}{14} \\
 &= 10 \text{ m s}^{-1}
 \end{aligned}$$

3

3 The diagram shows two opposite vertical forces of magnitude 1.2 N and 2.1 N acting